**LAB EXPERIMENT RECORD**

**SET 29: STUDENT PERFORMANCE AND PARTICIPATION ANALYSIS**

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|---|---|
| Department | AIDS |
| Course | DSA0206 / Data Handling and Visualization |
| Experiment | Data Visualization using R |
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## Aim

To visualize student test scores, participation scores, and study hours by attendance quartile using RStudio, and to describe an interactive dashboard for exploring student performance.

## Datasets

Dataset A — Student test and participation scores

| Student ID | Test Score | Participation Score |
|---|---|---|
| S1 | 85 | 8 |
| S2 | 70 | 7 |
| S3 | 92 | 9 |
| S4 | 80 | 8 |
| S5 | 65 | 6 |

Dataset B — Study hours and attendance quartile (as shown in the supplied screenshot)

| Student ID | Study Hours | Attendance Quartile |
|---|---|---|
| S1 | 8 | Q1 |
| S2 | 10 | Q2 |
| S3 | 15 | Q4 |
| S4 | 12 | Q3 |
| S5 | 7 | Q1 |

## PROGRAM 1: Density Plot of Test Scores

Objective: Display the distribution of student test scores using a kernel density estimate and rug marks.

```
data <- data.frame(Student_ID = paste0("S", 1:5),
                   Test_Score = c(85, 70, 92, 80, 65))
plot(density(data$Test_Score),
     main = "Density Plot of Test Scores",
     xlab = "Test Score", ylab = "Density",
     col = "blue", lwd = 2)
polygon(density(data$Test_Score), col = "lightblue",
        border = "blue")
rug(data$Test_Score, col = "red", lwd = 2)
print(data)
```

Procedure: Create the data frame, compute the kernel density estimate, draw the filled density curve, and add rug marks at the observed test scores.

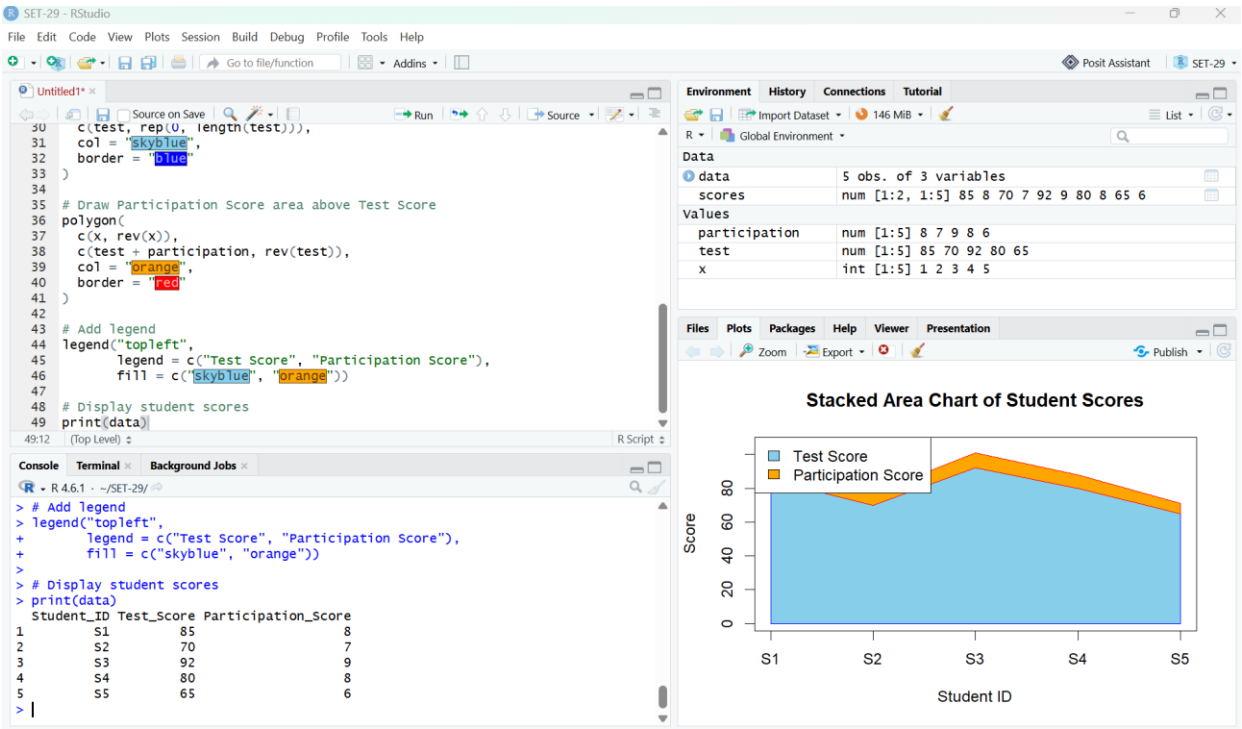



*Figure 1. RStudio density plot of test scores.*

Observation: Test scores range from 65 to 92. The density curve is a smoothed representation of five scores; it should not be interpreted as a population distribution.

## PROGRAM 2: Stacked Area Chart of Test and Participation Scores

Objective: Compare student test scores and participation scores in a stacked area visualization.

```
data <- data.frame(Student_ID = paste0("S", 1:5),
                   Test_Score = c(85, 70, 92, 80, 65),
                   Participation_Score = c(8, 7, 9, 8, 6))
x <- 1:nrow(data)
test <- data$Test_Score
participation <- data$Participation_Score
plot(x, test + participation, type = "n",
     xaxt = "n", xlab = "Student ID", ylab = "Score",
     main = "Stacked Area Chart of Student Scores",
     ylim = c(0, max(test + participation) + 5))
axis(1, at = x, labels = data$Student_ID)
polygon(c(x, rev(x)), c(test, rep(0, length(test))),
        col = "skyblue", border = "blue")
polygon(c(x, rev(x)), c(test + participation, rev(test)),
        col = "orange", border = "red")
legend("topleft", legend = c("Test Score", "Participation Score"),
       fill = c("skyblue", "orange"))
print(data)
```

Procedure: Create both score vectors, plot the total heights, fill the test-score area from zero, then stack the participation area above the test scores.

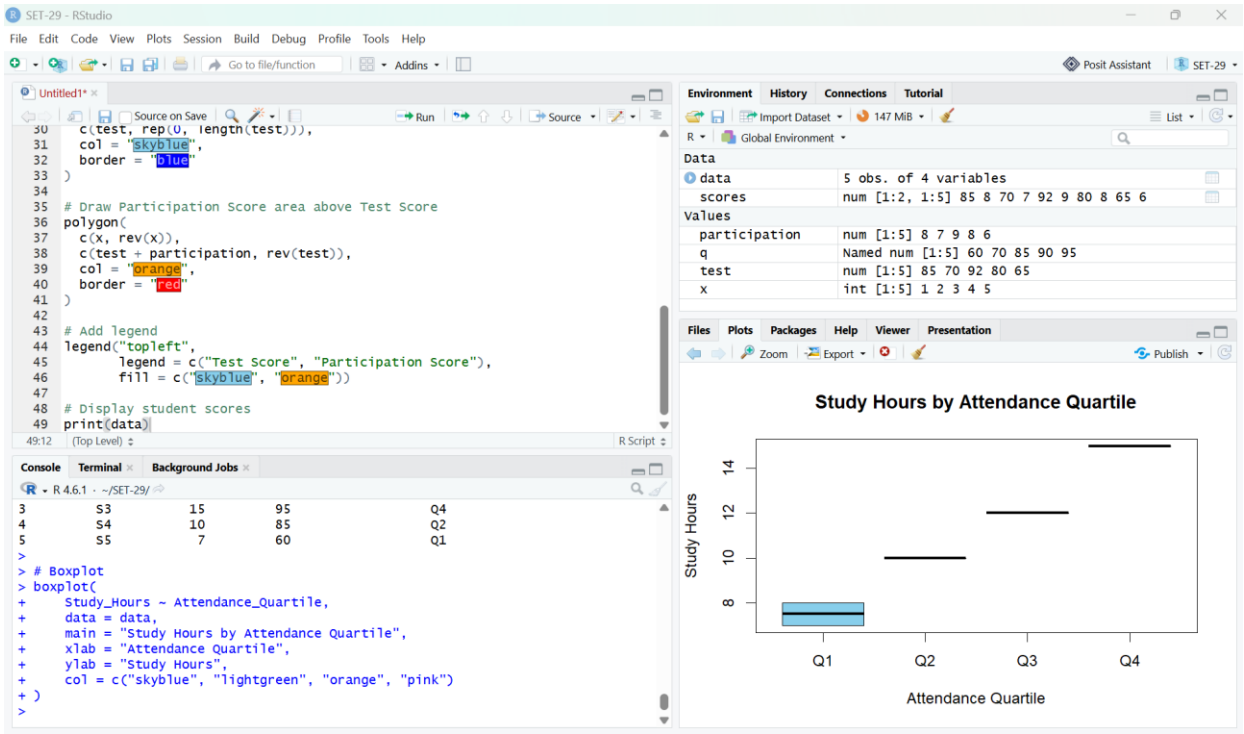



*Figure 2. RStudio stacked area chart of student scores.*

Observation: Student S3 has the highest test score (92) and participation score (9); S5 has the lowest (65 and 6). The two scores have different scales, so stacking is illustrative rather than a normalized comparison.

## PROGRAM 3: Boxplot of Study Hours by Attendance Quartile

Objective: Explore study hours grouped by attendance quartile.

```
data <- data.frame(
  Student_ID = paste0("S", 1:5),
  Study_Hours = c(8, 10, 15, 12, 7),
  Attendance_Quartile = factor(c("Q1", "Q2", "Q4", "Q3", "Q1"),
                               levels = c("Q1", "Q2", "Q3", "Q4"))
)
boxplot(Study_Hours ~ Attendance_Quartile, data = data,
        main = "Study Hours by Attendance Quartile",
        xlab = "Attendance Quartile", ylab = "Study Hours",
        col = c("skyblue", "lightgreen", "orange", "pink"))
print(data)
```

Procedure: Create the study-hours dataset, order quartile categories Q1 through Q4, and generate a grouped boxplot.

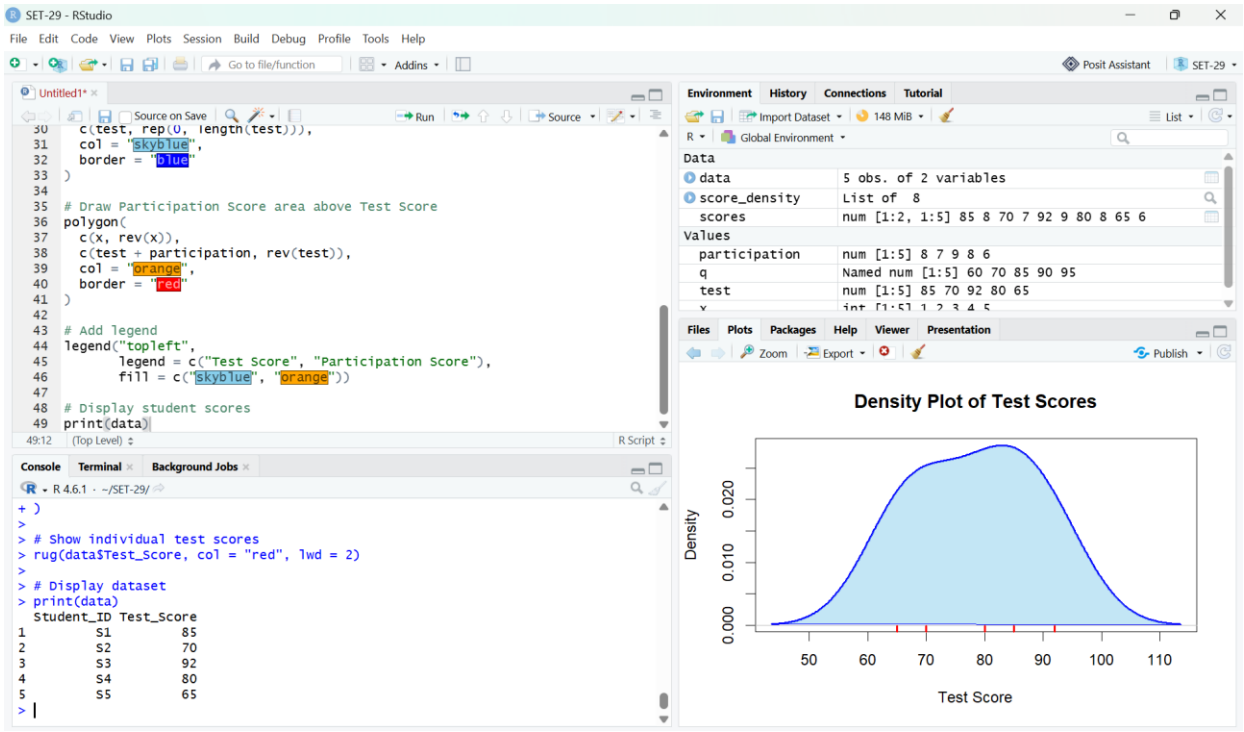



*Figure 3. RStudio boxplot of study hours by attendance quartile.*

Observation: Q1 contains two students studying 7 and 8 hours. Q2, Q3 and Q4 each contain only one student, so their boxplots appear as horizontal lines. A larger sample is needed to assess variability.

## PROGRAM 4: Interactive Dashboard (Tableau)

Procedure: Export the two datasets from R as CSV files; open them as separate Tableau data sources. Build a test-score view by Student_ID, a participation-score view, and a study-hours view by Attendance_Quartile. Arrange these worksheets on a dashboard and add Student_ID filters where the data sources are appropriately related.

```
write.csv(data.frame(Student_ID = paste0("S", 1:5),
                     Test_Score = c(85, 70, 92, 80, 65),
                     Participation_Score = c(8, 7, 9, 8, 6)),
          "SET29_Student_Scores.csv", row.names = FALSE)
write.csv(data.frame(Student_ID = paste0("S", 1:5),
                     Study_Hours = c(8, 10, 15, 12, 7),
                     Attendance_Quartile = c("Q1", "Q2", "Q4", "Q3", "Q1")),
          "SET29_Attendance.csv", row.names = FALSE)
```

Result: The Tableau dashboard procedure supports interactive exploration. No Tableau dashboard screenshot was supplied, so this section documents the steps rather than claiming it was executed.

## Overall Procedure

1. Enter the student datasets into RStudio.
2. Run Program 1 and inspect the density plot.
3. Run Program 2 and inspect the stacked area chart.

4. Run Program 3 and inspect the grouped boxplot.
5. Export CSV files and build the Tableau dashboard using the provided instructions.

## Overall Result

The three supplied RStudio outputs demonstrate a test-score density plot, a stacked area chart of test and participation scores, and a boxplot of study hours by attendance quartile. These visualizations support exploratory analysis of the five student records.